Total Quality Management (GMP)

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Total Quality Management (GMP) (A09762)

Short Title:	Total Quality Management (GMP)
Faculty	Science and Computing
Department	Science
Cluster	Quality, Regulatory and Compliance
Author	BWHELAN
Credits	5

Level: Introductory

Description of Module / Aims

This module covers aspects of quality management including teamwork, continuous improvement, the philosophy of total quality management (TQM), the quality gurus: Deming, Crosby, Juran, training and problem-solving. The student will be able to identify a production-related technical problem, assist in its investigation and analysis, propose solutions to the problem, and monitor its effectiveness.

Programmes

`QUAL-0012`	Higher Certificate in Science in Good Manufacturing Practice and Technology (`WD_SGMPT_C`)	2 / 4 / M

Indicative Content

- The principles of Total Quality Management; the role of the customer, both internal and external; changing customer needs
- Definition of quality in TQM terms; the philosophies of Deming, Crosby and Juran
- Reduction in rework and scrap levels, elimination of defects in materials and products; development of suggestions for improvement
- Concepts of Lean Manufacturing; reduction in manufacturing and distribution lead times; preventative maintenance
- Reviewing the fundamental stages of the manufacturing processes via flow diagrams, identification of potential causes of product defects
- Systematic approach to problem solving and failure analysis; documenting corrective action, proposing improvements and monitoring effectiveness
- Teamwork, types of teams; cross-functional, project, self-managing; managing the change from traditional to TQM cultures
- The Seven Quality tools: Pareto, ishikawa, histograms, statistical process control, process capability, checksheets, trending of data; problem-solving techniques

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain the concepts of Total Quality Management and Continuous Improvement.
2. Discuss the principles of variation in streamlining manufacturing process.
3. Review a production process and identify critical production bottlenecks.
4. Apply the appropriate techniques to problem solving in the manufacturing environment.

Learning and Teaching Methods

- Lectures.
- Case studies.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4
Continuous Assessment	50%	1,4
Essay	25%	1
In-Class Assessment	25%	4

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Independent Learning		111

Essential Material

- "The 7 Basic Quality Tools." http://www.thequalityweb.com/
- George, M.L., D. Rowlands, M. Price and J. Maxey. _Lean Six Sigma Pocket Toolbook_. New York: Mc-Graw Hill, 2005.
- Juran, J.M. _Juran's Quality Handbook_. London: Mc-Graw Hill, 2010.

Supplementary Material

- Oakland, J.S. _Statistical Process Control_. Oxford: Butterworth Heinmann, 2005.